

COURSE DESCRIPTION			
Program	Artificial Intelligence, Artificial Intelligence & Machine Learning, Cloud Computing and Big Data, Cyber Forensics and Information Security, Computer Science and Technology, Computer Science & Engineering, Computer Engineering, Computer Science and Engineering (Artificial Intelligence & Machine Learning), Food Processing Technology, Information Technology, Polymer Technology, Printing Technology, Automation and Robotics, Robotic Process Automation, Textile Technology		
Course Title	Fundamentals of Electrical and Electronics Engineering Lab		
Course Code	2138	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

The Fundamentals of Electrical & Electronics Engineering Lab provides students with basic practical knowledge of electrical and electronic engineering. The course helps students understand domestic electrical wiring, electrical safety, and the use of measuring instruments to measure voltage, current, power, and energy. Students also gain hands-on experience in identifying and testing electronic components, using electronic instruments, and studying rectifier circuits and logic gates. This laboratory course emphasizes learning by doing, develops practical skills, and supports foundational engineering learning.

Course Objectives

1	To provide hands-on experience in basic electrical wiring practices and electrical safety.
2	To enable students to use electrical measuring instruments to measure voltage, current, power, and energy.
3	To help students identify and test basic electronic components and understand their operation.
4	To introduce students to rectifier circuits and basic digital logic gates through practical experiments.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic electrical circuits; passive and active components (concurrent with the semester).	2131	Fundamentals of Electrical & Electronics Engineering	II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Construct basic domestic electrical wiring and lighting circuits by applying standard wiring practices and safety rules.	Apply
CO2	Measure electrical voltage, current, power, and energy in single-phase circuits using appropriate electrical measuring instruments.	Apply
CO3	Test and examine passive and active electronic components and electronic instruments to analyze their basic operating characteristics.	Apply
CO4	Implement rectifier circuits and logic gate circuits to verify their operation using waveforms and truth tables.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	2	3	-	-	-	-	-	-
CO2	3	2	-	2	3	1	-	-	-	-	-
CO3	3	2	2	2	3	-	-	-	-	-	-
CO4	3	2	2	2	3	-	-	-	-	-	-
Total	12	8	6	8	12	1	-	-	-	-	-
Correlation Level	3	2	2	2	3	1	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Familiarization	Familiarization of Domestic Electrical Wiring Materials and Standards.	CO1	Understand	3
2	Domestic Wiring.	Demonstrate the wiring of a Single-Phase Domestic Lighting Control Circuit using Source, Fuse, SPST Switch, and Lamp. Demonstrate the wiring of Staircase Lighting Circuit Using Two-Way Switches.	CO1	Apply	3
3	Voltage Division and Current Division Rule	Demonstrate and verify the Voltage Division rule in a Series Circuit containing different power rated Lamps. Demonstrate and verify the Current Division rule in a Parallel Circuit containing different power rated Lamps.	CO1	Apply	3
Module II					
4	Electrical Power Measurement	Demonstrate the usage of Wattmeter to measure electrical power in a single phase circuit.	CO2	Apply	3
5	Electrical Energy Measurement	Demonstrate the usage of an energy meter to measure electrical energy consumption in a single phase electrical circuit.	CO2	Apply	3
6	Extension Board	Wiring and Assembly of an Extension Board with Fuse, Sockets and Switches	CO2	Apply	3
	Internal Exam I		CO1	Apply	3
Module III					
7	Laboratory Instruments	Demonstrate various types of electronic instruments - ammeters, voltmeters, Digital multimeters, function generators, power supply and CRO/DSO	CO3	Understand	3
8	Testing of components using Multimeter	Testing of Passive and Active Electronic components and familiarization of bread board	CO3	Apply	3
9	Measurement of Signal Parameters	Measurement of amplitude, time period and frequency values of a sine wave using CRO/DSO	CO3	Apply	3
Module IV					
10	Half Wave rectifier and Center tapped rectifier	Construct half wave rectifier with and without filter and plot the input / output waveforms and calculate the ripple factor. Construct center tapped rectifier with and without filter and plot the input / output waveforms and calculate the ripple factor.	CO4	Apply	3
11	Bridge rectifier	Construct a bridge rectifier with and without filter and plot the input / output waveforms	CO4	Apply	3
12	Logic Gates	Examine the truth table of digital logic gates.	CO4	Apply	3
	Internal Exam II		CO4	Apply	3
	Open Ended Experiment		CO4	Apply	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments. Any 3 to 5 open ended problems may be chosen and divided among student groups.

Students may choose to:

Choose an open ended project based on one or more of the following aspects:

1. Modify or extend an existing experiment to explore additional parameters or behaviors.
2. Integrate multiple concepts from the syllabus into a combined application
3. Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
4. Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Describe why the switch is connected in the phase line. What will happen if the switch is connected in the neutral line instead of the phase line?	1.5
2	Identify various electrical safety mechanisms available in your home.	1.5
3	Explain the relationship between lamp power rating, voltage drop, and brightness in a series circuit	1.5
4	Identify whether domestic electrical appliances are connected in series or parallel. Explain why this arrangement is used.	1.5
Module II		
5	List various electrical appliances used in your home. Identify their power ratings and classify them as high-power and low-power devices.	1.5
6	Identify the energy star rating of an electrical appliance in your home and explain what this rating reveals about its energy efficiency.	1.5
7	Record the initial and final energy meter readings of your home for a chosen time period. Calculate the number of units consumed and estimate the electricity bill for that period. Compare it with the energy bill given by the electricity board.	1.5
Module III		
8	Prepare a chart containing the passive and active electronic components with their diagrams and symbols with proper labeling	1.5
9	Prepare a list of various types of diodes with their key features.	1.5
10	Create a chart illustrating the step-by-step multimeter testing procedures for a diode and a transistor with diagrams.	1.5
11	Collect the datasheet of any one BJT. Draw its pin diagram and prepare a note on its key specifications such as voltage ratings, current rating, power dissipation, and gain.	1.5
Module IV		
12	Prepare a chart showing the comparison of various parameters of Half wave rectifier, Full wave Center tapped rectifier and Full wave bridge rectifier.	1.5
13	Prepare a table showing the difference between conventional linear regulated power supply and SMPS	1.5
14	Prepare a list of various logic families and write one key feature of each.	1.5
15	Draw the diode-logic implementation of AND and OR gates and verify their truth tables	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Electronics Lab Manuals	K A Navas	PHI
2	Electronics Lab Manuals	Syammohan S, Kuriachan D	Ayodya

REFERENCE			
Sl. No.	Title	Author	Publication
1	Basic Electronics: A Text-lab Manual	Paul B. Zbar, Albert Paul Malvino, Michael A. Miller	McGraw-Hill Publishing Company

WEB RESOURCES			
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Sl. No.	URL	Modules Covered
1	https://www.digimat.in/nptel/courses/video/108105112/L01.html	I & II
2	https://www.digimat.in/nptel/courses/video/108101091/L01.html	III & IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Digital Storage Oscilloscope	3 MHz, Dual channel	15
2	Signal/ Function Generator	3 MHz minimum	15
3	Power Supply Unit	30V, Dual	15
4	Multimeter	DMM	15
5	Ammeter	0-2A	15
6	Voltmeter	0-230V	15
7	Energy Meter	230V, 50Hz	15
8	Digital Trainer Kit		15

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to			15	15	10	
Marks	5	30			10	40
Total marks			60			40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10

Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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